Trygve Haavelmo and the Simultaneous-Equations Problem

Haavelmo's central contribution was to show that simultaneously determined economic variables cannot in general be analysed with ordinary single-equation regression. The equations must instead be understood as parts of a joint probabilistic system.

A market for one good

Write demand and supply in logs, so that the slopes are elasticities:

$$Q_t^d = \alpha_0 + \alpha_1 P_t + u_{dt}, \qquad \alpha_1 < 0,$$

$$Q_t^s = \beta_0 + \beta_1 P_t + u_{st}, \qquad \beta_1 > 0,$$

together with the equilibrium condition $Q_t^d = Q_t^s = Q_t$.

Price is not externally given. Solving the two equations for P_t:

$$P_t = \frac{(\beta_0 - \alpha_0) + (u_{st} - u_{dt})}{\alpha_1 - \beta_1}, \qquad \text{so} \qquad \frac{\partial P_t}{\partial u_{dt}} = \frac{1}{\beta_1 - \alpha_1} > 0.$$

A positive demand shock raises the equilibrium price. Consequently, taking u_{dt} and u_{st} as uncorrelated with variances σ_d^2 and σ_s^2,

$$\mathrm{Cov}(P_t, u_{dt}) = \frac{\sigma_d^2}{\beta_1 - \alpha_1} \neq 0.$$

Regressing quantity on price by OLS therefore does not recover α_1. Its probability limit is

$$\operatorname{plim} \hat{\alpha}_1^{OLS} = \frac{\alpha_1 \sigma_s^2 + \beta_1 \sigma_d^2}{\sigma_d^2 + \sigma_s^2},$$

a variance-weighted average of the two structural slopes. The observed price–quantity correlation reflects movements in both curves, not movement along one of them. This is "simultaneous-equations bias".

Note the limiting case: if demand is stable and only supply shifts ($\sigma_d^2 \to 0$), OLS does recover α_1, because supply shifts trace out the demand curve. That is identification through exogenous variation, in one line.

A distinction worth being precise about. OLS is consistent for $E(Q \mid P)$. The estimator is not the problem; the estimand is. $E(Q \mid P)$ is a feature of the observed joint distribution, and there is no reason for it to coincide with the structural demand relation.

Reduced form and identification

Add a supply shifter Z_t (weather, feed cost) that does not enter demand:

$$Q_t^s = \beta_0 + \beta_1 P_t + \gamma Z_t + u_{st}.$$

Solving the system gives the reduced form, in which each endogenous variable is expressed in terms of the exogenous variable and the shocks alone:

$$P_t = \pi_{10} + \pi_{11} Z_t + v_{1t}, \qquad Q_t = \pi_{20} + \pi_{21} Z_t + v_{2t}.$$

Here $\pi_{11} = \gamma/(\alpha_1 - \beta_1)$ and $\pi_{21} = \alpha_1 \pi_{11}$, so that

$$\alpha_1 = \frac{\pi_{21}}{\pi_{11}}.$$

The reduced form is always estimable by OLS. Identification is the question of whether the map from reduced-form parameters back to structural parameters can be inverted. Excluding Z_t from demand is exactly what makes it invertible here, and this ratio is the IV/indirect-least-squares estimator.

Haavelmo's argument, in four connected parts

1. The system, not the single equation, is the statistical model. Economic theory specifies the structural equations; assumptions about the disturbances give the system a joint probability distribution. That joint distribution determines how the observable data are generated.

2. Structural equations are not regressions. A demand equation describes a behavioural or causal relationship, what happens to quantity demanded when price changes, holding the relevant structural conditions fixed. It is not in general the conditional expectation observed in the data.

3. Identification. The joint distribution of observed prices and quantities may be consistent with many different demand-and-supply structures. Structural parameters can be estimated only when theory supplies sufficient restrictions. The problem itself was not new, Working (1927), "What do statistical demand curves show?", and Frisch's confluence analysis had posed it earlier. Haavelmo's move was to state it inside a joint probability model; the general rank and order conditions came afterwards with Koopmans and the Cowles Commission.

4. A probabilistic basis for system estimation. Once the complete structural system and the distribution of its disturbances are specified, one can derive the probability distribution of the observed variables and construct likelihood-based estimators.

What he did not do

Haavelmo did not invent two-stage least squares. LIML is Anderson and Rubin (1949); 2SLS is Theil (1953) and Basmann (1957). He was not remote from application, though: Girshick and Haavelmo (1947) estimated a five-equation structural model of the demand for food. His achievement was more fundamental than any single estimator, he explained why such methods are necessary and supplied the probabilistic and identification framework on which they were built.

In modern terminology

$$\text{economic equilibrium} + \text{structural shocks} \quad \Longrightarrow \quad \text{endogenous regressors},$$

and causal effects require structural assumptions and identification, not merely correlations in observational data. Simultaneity is one source of endogeneity among several; measurement error, omitted variables and selection are others, and the same logic applies to each.

Sources and a note on chronology

The monograph was written before the article. “On the Theory and Measurement of Economic Relations” was completed as a hectographed manuscript in 1941. Haavelmo’s Harvard thesis, and appeared only in 1944, as a supplement to *Econometrica*, while he was working in New York for the Norwegian government-in-exile. He had been Frisch’s assistant in Oslo, and was professor at the University of Oslo from 1948 until 1979.

- Haavelmo, T. (1943). “The Statistical Implications of a System of Simultaneous Equations.” *Econometrica* 11(1), 1–12. https://www.jstor.org/stable/1905714
- Haavelmo, T. (1944). “The Probability Approach in Econometrics.” *Econometrica* 12, Supplement, iii–115. https://www.jstor.org/stable/1906935
- Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel, 1989, awarded “for his clarification of the probability theory foundations of econometrics and his analyses of simultaneous economic structures.” https://www.nobelprize.org/prizes/economic-sciences/1989/press-release/
- Heckman, J. and Pinto, R. (2015). “Causal Analysis after Haavelmo.” *Econometric Theory* 31(1), 115–151, on the early and remarkably modern conception of causal intervention in Haavelmo’s structural equations. https://pmc.ncbi.nlm.nih.gov/articles/PMC4341827/